

A ReplayDr.GRPO (ours) – Dr.GRPO across three model scales

	Δ pass@8					
Qwen 0.5B	+0.65	+0.19	+0.36	+0.28	+0.21	+0.34
Falcon 1B	+0.15	+0.31 [†]	+0.98	+0.64	+0.12	+0.44 [†]
Qwen 3B	+0.15	+0.14	+0.59	+0.41	+0.11	+0.28
	Graph	Countdown	Python	MathIR	Pantry	Average

	Δ distinct@8					
Qwen 0.5B	+2.11	+1.15	+0.40	+0.32	+1.02	+1.00
Falcon 1B	+0.68	+0.42 [†]	+1.11	+0.66	+0.94	+0.77 [†]
Qwen 3B	+0.58	+0.33	+0.78	+0.43	+0.67	+0.56
	Graph	Countdown	Python	MathIR	Pantry	Average

B Initial reference + alternatives – terminal Dr.GRPO at Qwen2.5-0.5B

	Δ pass@8					
Before training	+0.13	-0.38	-0.17	-0.29	+0.38	-0.07
ReplayDr.GRPO (ours)	+0.65	+0.19	+0.36	+0.28	+0.21	+0.34
MaxRL (no replay)	+0.21	+0.10	+0.00	-0.07	+0.01	+0.05
UCPO	-0.00	+0.13	+0.00	+0.03	+0.05	+0.04
RLEP-Dr	+0.10	+0.10	+0.00	+0.13	+0.00	+0.07
Semantic-MaxEnt	+0.00	+0.14	+0.00	-0.12	+0.20	+0.04
GRPO	+0.04	+0.16	+0.00	-0.05	+0.02	+0.03
	Graph	Countdown	Python	MathIR	Pantry	Average

	Δ distinct@8					
Before training	+0.24	-0.38	-0.17	-0.29	+1.90	+0.26
ReplayDr.GRPO (ours)	+2.11	+1.15	+0.40	+0.32	+1.02	+1.00
MaxRL (no replay)	+0.35	+0.13	+0.00	-0.07	+0.01	+0.09
UCPO	-0.00	+0.18	+0.00	+0.03	+0.05	+0.05
RLEP-Dr	+0.15	+0.13	+0.00	+0.12	+0.00	+0.08
Semantic-MaxEnt	+0.00	+0.18	+0.00	-0.12	+1.12	+0.24
GRPO	+0.05	+0.17	+0.00	-0.05	+0.02	+0.04
	Graph	Countdown	Python	MathIR	Pantry	Average